

Stellar Classification Using Machine Learning Approaches

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Abstract

Classifying stars by spectral type is critical in astrophysics for determining temperature, gravity, and metallicity. Astrophysical classification is the categorization of celestial objects based on their fundamental physical properties. This approach reveals key parameters such as temperature, surface gravity, and metallicity, enabling us to understand their evolutionary processes. Traditional methods often rely on extensive spectral data, motivating more efficient computational approaches. We evaluated a fully connected neural network (FCNN) that integrates Linear Discriminant Analysis (LDA) for dimensionality reduction, as well as batch normalization, dropout, and a class-weighted loss function for handling imbalanced data. On a smaller dataset, the model attained an accuracy that effectively captured key patterns. When trained on a larger dataset, it maintained a high accuracy, indicating scalability. Accurate spectral classification also provides important insights into stellar populations, star formation histories, and the chemical evolution of galaxies, enhancing our broader understanding of astrophysical processes. However, rare classes (e.g., O-type stars) remain challenging due to their scarcity. SHAP analyses revealed that photometric magnitudes (G (mag), RP (mag), and BP (mag)) and effective temperature T_{eff} (K) play crucial roles in the model's predictions. While these results highlight the potential for rapid, automated stellar classification, future research could explore more advanced architectures or hybrid methods to improve performance on underrepresented classes.

1 Introduction

Stellar classification is essential in astrophysics, enabling astronomers to categorize stars based on observable properties. This system facilitates the study of stellar formation, evolution and underlying physical processes. Originating in the early 20th century, the Harvard Spectral Classification system categorizes stars by surface temperature and spectral characteristics, introducing classes O, B, A, F, G, K, and M [1].

The Morgan-Keenan (M-K) System refines this by incorporating luminosity classes (I for supergiants, II for bright giants, III for normal giants, IV for subgiants, and V for main sequence) [2]. Additionally, we incorporate predictive steps within the Hertzsprung–Russell (HR) diagram to confirm whether our star classifications align with established evolutionary tracks, offering a straightforward interpretive check.

2 Data

At the beginning of this project, sample datasets from various sources such as Kaggle, NASA, and LAMOST were used, and the initial model was tested with these samples. Although these datasets included almost all star classes based on the Harvard Spectral Classification (OBAFGKM), they lacked a sufficient number of parameters and were relatively small in size. Therefore, in order to make a more comprehensive classification, larger and more extensive datasets were required.

To address this need, we reached out to LAMOST to obtain the most suitable data set for this project and received the final dataset from the LAMOST Na-

tional Astronomical Observatories, Chinese Academy of Sciences. This dataset included the latest records for all star classes and contained a total of 10 million star records (~3GB) with 40 parameters (features) and 63 subclasses, making this dataset sufficient for this project [3].

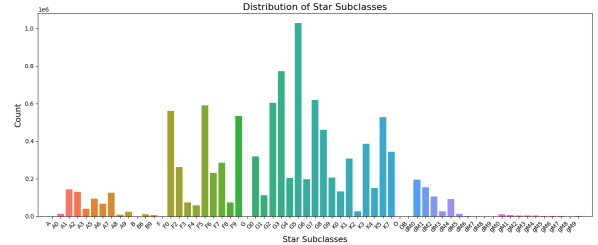


Figure 1: Distribution of subclasses in the dataset.

In this project, several data preprocessing steps were applied to prepare the dataset for model training. Since we had enough data (except for the O and B-class), rows with missing data were removed to ensure consistency. Only numerical features were selected, and the data was standardized using `StandardScaler` to bring all variables to a similar scale. Categorical target labels were encoded as numerical variables using `LabelEncoder` for compatibility with the model.

To address class imbalance, particularly the scarcity of O and B-type stars (see Figure 1), the Synthetic Minority Oversampling Technique (SMOTE) [4] was initially considered to generate synthetic samples. However, a decrease in model performance was observed when using SMOTE, so it was discarded. Instead, the natural dataset was used, preserving the original class distribution.

The parameters T_{eff} (K) (effective temperature), $\log g$ (dex) (surface gravity), and $[\text{Fe}/\text{H}]$ (metallicity) are among the most critical features for star classification, providing essential insights into a star’s spectral type, evolutionary stage, and chemical composition [5]. T_{eff} (K) defines spectral types (OBAFGKM); hot O-type stars ($T_{\text{eff}} > 30,000$ K) emit ultraviolet light, while cooler M-type stars ($T_{\text{eff}} < 3,500$ K) emit infrared radiation. Additionally, T_{eff} categorizes stars based on radiation output, revealing details about their energy production and age [6].

Logarithmic scales are often employed for parameters like $\log g$ (dex) and $[\text{Fe}/\text{H}]$, commonly denoted as (dex) in datasets, to compress wide value ranges and facilitate meaningful comparisons [7]. $\log g$ distinguishes between main-sequence stars (with higher surface gravity) and giant stars (with lower surface gravity), while $[\text{Fe}/\text{H}]$ reflects a star’s metallicity and age, associating metal-poor stars with the galactic halo and metal-rich stars with younger stellar populations [8].

Building on this foundation, the dataset provided by LAMOST aligns with the M-K classification system and includes 63 subclasses. Since using all subclasses in the initial training would be computationally expensive, we employed the 7-class Harvard Spectral Classification (O, B, A, F, G, K, M), which saved computational resources while preserving core stellar properties.

Class	Count
A	641,602
B	18,069
G	4,521,825
F	2,664,077
K	1,873,056
M	625,161
O	243

Table 1: Distribution of the 7 Harvard classes in the dataset.

Our initial tests revealed that using only T_{eff} (K) as the classification feature resulted in significantly lower accuracy. To improve performance, we incorporated additional parameters such as $\log g$ (dex), $[\text{Fe}/\text{H}]$, color indices, and magnitudes, aiming for a more robust classification process that better captures the complexity of stellar properties.

3 Astrophysical Metrics

Astrophysical metrics provide deeper insight into stellar evolution and properties. Below, we briefly outline two key formulations. These metrics often appear in the Hertzsprung–Russell (H-R) diagram, which maps stars by luminosity and temperature.

Table 2: Selected Astrophysical Metrics and Their Data Types

Metric	Data Type	Description
Teff (K)	Continuous	Effective temperature (K)
log g (dex)	Continuous	Surface gravity
ci	Continuous	Color index (BP - RP)
absmag	Continuous	Absolute magnitude
class	Categorical	Spectral class label

3.1 Mathematical Formulations

3.1.1 Color Index (ci)

$$\text{ci} = \text{BP} - \text{RP}. \quad (1)$$

A lower or negative **ci** indicates a hotter star, while a higher **ci** suggests a cooler star.

3.1.2 Absolute Magnitude (absmag)

$$\text{absmag} = G - 5 \times \log_{10}(\text{r_geo}) + 5. \quad (2)$$

Lower **absmag** values correspond to more luminous stars. Here, G is the apparent magnitude in a specific band, and r_geo is the geometric distance in parsecs.

4 Hertzsprung–Russell Diagram

4.1 Traditional HR Diagram

The Hertzsprung–Russell (HR) diagram correlates stellar luminosities with surface temperatures, revealing evolutionary stages such as the main sequence, giants, and supergiants [9, 10]. This diagram underscores the fundamental link between a star’s energy output and its photospheric temperature, indicating where it stands in its lifecycle. By plotting many stars, astrophysicists can discern mass-luminosity relationships, identify sub-populations (e.g., subgiants, horizontal branch stars), and examine stellar birthrates across different galactic regions [11, 12].

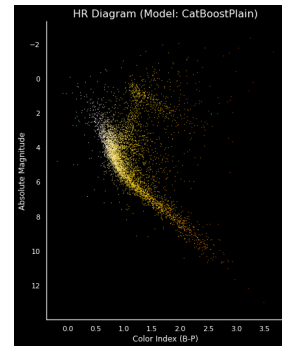


Figure 2: Hertzsprung–Russell Diagram showing the distribution of stars based on luminosity and temperature.

4.2 Metallicity in the HR Diagram

Metallicity ($[\text{Fe}/\text{H}]$) modifies a star’s evolution by affecting its opacity, energy transport, and fusion processes [8].

Higher metallicities often lead to increased line blanketing, causing stars to have cooler surface temperatures for a given luminosity. Metal-poor stars may remain hotter or dimmer, influencing overall lifetimes and evolutionary outcomes.

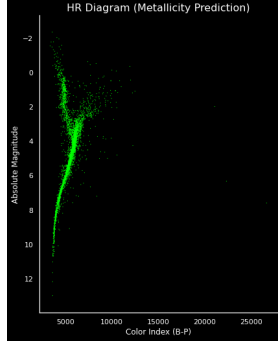


Figure 3: Metallicity Distribution in the Hertzsprung-Russell Diagram illustrating the impact of $[\text{Fe}/\text{H}]$ on star classification.

5 Linear Discriminant Analysis (LDA)

LDA is a dimensionality reduction technique that projects data onto a lower-dimensional space to maximize class separability for classification tasks [13]. It can be used to create new features or directly for classification purposes [14].

Within-class Scatter Matrix : $S_W = \sum_{i=1}^C \sum_{x \in C_i} (x - \mu_i)(x - \mu_i)^T$ [13] (3)

Between-class Scatter Matrix : $S_B = \sum_{i=1}^C N_i(\mu_i - \mu)(\mu_i - \mu)^T$ [13] (4)

Maximum number of LDA Components = $\min(d, C - 1)$ (5)

where:

- d : Dimensionality of the input features.
- C : Number of independent directions required to distinguish between C classes.

How it Works

- **Focus on Separation:** LDA projects data onto a lower-dimensional space where the separation between different classes is maximized.
- **Main Goal:** It aims to find the best linear combinations of the original features to create a new set of features.
- **Key Idea:** The transformed features maximize the distance between the means of classes while minimizing the variation within each class [13].

After solving the eigenvalue and eigenvector problem using the optimized scatter matrices S_W and S_B , the LDA coefficient matrix (W) is obtained.

$$S_W^{-1} S_B W = \lambda W \quad (6)$$

LDA Transformation : $X_{LDA} = X_{scaled} \cdot W$ (7)

Posterior Probability Matrix from the Model : $P(c_i | X_{LDA}) = f(X_{LDA}) = \hat{Y}$ (8)

Class Prediction : $Y_{Pred} = \text{argmax}(\hat{Y})$ (9)

Challenges

- **Dimensionality Reduction:** LDA reduces dimensionality, which may result in the loss of critical information, especially when dealing with complex datasets.
- **Underrepresented Classes:** Classes with a small number of samples in the dataset can pose challenges, as LDA might struggle to adequately represent these underrepresented classes [13].

6 Methods

6.1 Deep Learning

Deep learning, a sub-branch of machine learning, uses ANN technology, inspired by the human brain, to learn and make decisions from large amounts of data. Unlike traditional machine learning methods, it treats raw data as direct input and automatically extracts meaningful features from this data thanks to its layered structure.

This computational progression through the network is called forward propagation. The input and output layers of a deep neural network are called visible layers. The input layer is where the deep learning model receives data for processing and the output layer is where the final prediction or classification is made [15].

Another process called back-propagation uses algorithms such as gradient descent to calculate the errors in the predictions and then adjusts the weights and biases of the function by moving backwards through the layers to train the model. Together, forward propagation and back propagation enable a neural network to make predictions and correct errors. Over time, the algorithm becomes increasingly accurate [15].

6.2 Model Selection Process

Choosing the right model for star classification is critical to ensure accuracy and reliability in results. Recent research has examined several approaches, each designed to address unique challenges, such as gradient boosting, random forests and various deep learning models.

Each model has been tested individually to evaluate its performance. Most of the traditional models achieved

close to 80% accuracy, and When the f1-score, which is the harmonic mean of precision and recall, is evaluated, it is clear that the models are inadequate in predicting the type O stars. Based on these results, ensemble methods such as voting and stacking were applied. However, contrary to expectations, these methods resulted in lower accuracy rates and did not meet the desired performance.

To achieve better results, more advanced models were tested. Among them, Fully Connected Neural Networks (FCNNs) performed the best, reaching 84% accuracy. Therefore, we decided to continue our study using the FCNN model.

Table 3: Classification Performance

Class	Accuracy	Recall	F1-Score
Catboost	0.82	0.82	0.82
Lightboost	0.80	0.80	0.80
XGboost	0.82	0.82	0.82
Multi layer perceptron	0.81	0.81	0.80
Votting(MLP+Catboost)	0.72	0.72	0.72
NCF	0.80	0.80	0.80
FCNNs	0.84	0.84	0.84

6.3 StarDeepNN

Architecture

In this paper, we address the star classification problem with a PyTorch-based Fully Connected Neural Network (FCNN) model.

- **Input Layer:** Accepts the dimension-reduced data from LDA.
- **Hidden Layers:**
 1. 256 neurons with Batch Normalization, ReLU activation, and 40% Dropout.
 2. 128 neurons with Batch Normalization and ReLU activation.
 3. 64 neurons with Batch Normalization and ReLU activation.
- **Output Layer:** 7 neurons, suitable for multi-class classification with the Softmax function.

Batch Normalization and Dropout

Degradation problems arise when deep models start converging: with the network depth increasing, accuracy can get saturated and then degrade rapidly. This problem may be caused by back-propagation gradients that vanish or explode by the time they reach the earlier layers [16].

While there are various methods to avoid this situation, the combination of batch normalization and dropout provides an effective solution.

CrossEntropyLoss and Optimization

PyTorch’s CrossEntropyLoss feature calculates the difference between the raw outputs (logits) of the model and the true labels. Mathematically, it is expressed as follows [17]:

$$l_n = -w_{y_n} \log \left(\frac{\exp(x_{n,y_n})}{\sum_{c=1}^C \exp(x_{n,c})} \right) \quad (10)$$

For imbalanced datasets, class weights allow the model to focus more on underrepresented classes. Therefore, we calculated the weight of each class to be inversely proportional to the number of data instances of that class, giving higher weights to classes with fewer instances so that the model learns these classes better.

The weights of the model are updated using the Adam optimization algorithm. Adam provides an efficient training process by setting an adaptive learning rate for each parameter. Weight decay (L2 regularization) was added to prevent the weights from growing too large, thus increasing the generalization capacity of the model. As noted in related studies,[18] weight decay reduces the scale of the weights, thereby increasing the effective learning rate and enhancing the regularization effect of gradient noise. This mechanism helps prevent overfitting by limiting the size of the weights and promoting better generalization.

7 Shapley Additive Explanations (SHAP)

SHAP is an explainability method that assigns each feature a value indicating its contribution to a model’s prediction, based on Shapley values from game theory [19].

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{\text{Weight for Subset } S}{\text{Total Subsets}} \cdot [f(S \cup \{j\}) - f(S)] \quad [19] \quad (11)$$

- S : A subset of features excluding feature j .
- $f(S)$: The model’s prediction using only features in subset S .
- ϕ_j : The importance of the j -th feature.

Feature Contributions to LDA Components

$$\phi_{LDA}^{C_i} = \text{SHAP}(f, X_{LDA}) \quad (12)$$

$$\phi_{\text{original}}^{C_i} = \phi_{LDA}^{C_i} \cdot W_{LDA}^T \quad (13)$$

$$\phi_{\text{overall}} = \frac{1}{C} \sum_{i=1}^C \left| \phi_{\text{original}}^{C_i} \right| \quad (14)$$

- $\phi_{LDA}^{C_i}$: The SHAP values computed in the LDA space for a class C_i

- $\phi_{\text{original}}^{C_i}$: SHAP values computed in the original feature space for a class C_i
- ϕ_{overall} : The average SHAP value of a feature across all classes, highlighting its overall importance in the model.

SHAP Analysis

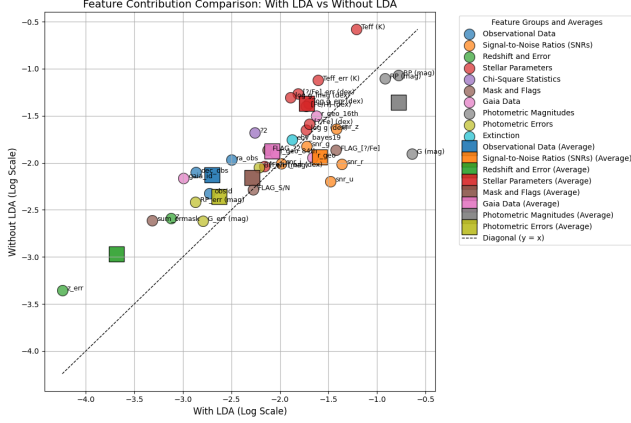


Figure 4: Feature contributions with vs. without LDA, highlighting changes in importance across feature groups on log scale.

Points Above the Diagonal (With LDA < Without LDA):

These points indicate that the feature's contribution decreases when LDA is applied, meaning LDA reduces its importance.

Points Below the Diagonal (With LDA > Without LDA):

These points show that the feature's contribution increases when LDA is applied, meaning LDA enhances its importance.

Points on the Diagonal (With LDA $\approx$ Without LDA):

These points suggest that there is no significant change in the feature's contribution before and after applying LDA.

Without LDA

- Features from the **Stellar Parameters** group, such as T_{eff} (K), $[\alpha/\text{Fe}]$, and $[\text{Fe}/\text{H}]$, dominate the model's decisions.
- Features from the **Photometric Errors** group, such as $T_{\text{eff_err}}$ (K) and $\log g$ (dex), contribute minimally.
- Features from the **Photometric Magnitudes** group, such as G (mag), RP (mag), and BP (mag), have less impact.

With LDA

- Features from the **Photometric Magnitudes** group, such as G (mag), RP (mag), and BP (mag), become the most impactful.
- Features from the **Stellar Parameters** group, such as T_{eff} (K) and $\log g$ (dex), lose significance.

The use of LDA amplifies differences in **Photometric Magnitudes** (e.g., G (mag), RP (mag), and BP (mag)) while de-emphasizing contributions from **Stellar Parameters**

8 Results and Discussion

On the smaller dataset, the model achieved an accuracy of 86.70%, demonstrating its ability to capture key patterns in the data. When trained on a larger dataset, the model maintained an accuracy of 84%, demonstrating its scalability. However, classification for rare classes remains a challenge, with limited performance for the underrepresented O class.



Figure 5: Confusion matrix and classification report

The SHAP analysis provided valuable insights into the model's decision-making process by identifying key features such as RP (mag), BP (mag), G (mag) and T_{eff} (K) as the most effective predictors. This is not only in line with astrophysical expectations, but also improves the interpretability of the model, making it more transparent and reliable for experts in the field.

While the results are promising, some challenges remain. For example, all models had problems predicting class O due to its scarcity in the data and its similarity to class B. Although this problem was relatively reduced

when the weights were changed, this led to a decrease in the overall performance of the model.

Moreover, the complex nature of stellar spectra requires exploration of more advanced model architectures or hybrid methodologies that can capture subtle patterns and dependencies, paving the way for more accurate and reliable predictions.

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